

Course Title: Exploratory Data Analysis using Python

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Project Overview

The goal of this capstone project is to give you an immersive, hands-on opportunity to apply all the skills and concepts you've learned throughout the course—using Python, not just as a coding tool, but as a powerful language for exploring, analyzing, and communicating insights from real-world data.

Working with authentic, publicly available datasets, you will demonstrate your ability to extract, clean, transform, and analyze data using Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Plotly. You'll also integrate GenAI tools like ChatGPT or Gemini to refine problem statements, generate summaries, and support storytelling through natural language.

This project brings together all the core competencies built across the sprints—data preparation, EDA, visualization, dashboard creation, narrative crafting, and presentation skills—into one cohesive, outcome-driven experience. You'll be challenged to think like an analyst: identify a business-relevant problem, explore messy data, extract patterns and insights, and communicate them clearly through visuals and narratives.

Your task is to:

1. **Select a topic** from the suggested list or propose your own with mentor approval.
2. **Collect/download** a relevant dataset from trusted, public data sources.
3. **Extraction, clean, transform** and **analyze** the dataset using Python (Pandas, NumPy, etc.).
4. **Apply univariate, bivariate, and multivariate analysis techniques** to uncover meaningful patterns and trends.
5. **Use GenAI tools** to craft/refine EDA questions, generate summaries, and assist in presenting your insights
6. **Create a compelling interactive dashboard** using **Streamlit** or **Plotly Dash** that brings your findings to life.
7. **Document your work** in a structured format including project plan, markdown summaries, and presentation slides.
8. **Present your insights** in a 3–4 minute pitch using your dashboard and visuals—just like a real-world analyst would!

This capstone is your moment to shine—to connect the dots, show your thinking, and walk away with a portfolio-worthy project that demonstrates your readiness for future academic, internship, or professional opportunities.

Project Objectives

By completing this capstone project, you will be able to:

- **Apply essential data analysis techniques** to extract, clean, transform, and summarize data using Python libraries such as **Pandas** and **NumPy**.
- **Uncover insights from real-world datasets** through structured **univariate, bivariate, and multivariate analysis**, supported by descriptive statistics and visual exploration.
- **Create interactive, visually engaging dashboards** using tools like **Matplotlib, Seaborn, Plotly**, and **Streamlit** that clearly communicate your findings.
- **Tell compelling data stories** by integrating visuals, narratives, and analytical insights into a cohesive and professional presentation format.

- **Use GenAI tools** (like **ChatGPT** and **Gemini**) to refine business questions, summarize insights, generate markdown documentation, and enhance storytelling.
- **Work collaboratively and responsibly** by planning tasks, delegating roles, sharing files efficiently, and maintaining smooth team communication.
- **Strengthen your project execution abilities** by setting clear goals, defining milestones, and delivering outcomes with quality and consistency.
- **Demonstrate confidence and readiness** to apply your Python-based EDA skills in academic, professional, or portfolio-building contexts.

This project aims to not only assess your technical competency but also elevate your ability to solve problems, communicate insights, and collaborate effectively—just like data analysts do in the real world.

Project Workflow and Implementation Steps

Follow these structured steps to plan, build, and present your capstone project effectively using **Python-based EDA techniques and tools**.

1. Select a Topic of Interest

- Choose one topic from the suggested list or propose your own (with mentor approval).
- Ensure the topic allows for meaningful data exploration and insight-driven storytelling.

2. Data Collection

- Search and download real-world datasets from reliable sources (e.g., Kaggle, govt portals, UCI, etc.).
- Ensure the dataset is relevant, well-structured, and ethically sourced.
- Validate that the data supports your project questions and can yield multiple layers of insight.

3. Data Import and Initial Exploration

- Import the dataset(s) into your Python environment (Jupyter Notebook or Google Colab).
- Use Pandas to explore the dataset: structure, column types, shape, null values, and data quality indicators.

4. Data Cleaning and Transformation

- Handle missing values, fix data type mismatches, remove duplicates, and normalize values.
- Rename columns, split/merge fields, filter rows, and format data using Pandas and NumPy.
- Optionally store cleaned data in a lightweight database (e.g., SQLite or MySQL) for structured access.

5. Exploratory Data Analysis (EDA)

- Apply univariate, bivariate, and multivariate analysis using Pandas, Seaborn, Plotly, and Matplotlib.
- Use techniques like grouping, aggregating, pivoting, correlation matrices, and visual summaries.
- Detect patterns, trends, and outliers to support your business questions.

6. Use GenAI for Insight Framing (Optional but Recommended)

- Use tools like ChatGPT or Gemini in Colab to refine your business problem and questions.
- Ask GenAI to generate markdown explanations, summaries of trends, or narration for your dashboard.
- Use GenAI as a co-pilot, not a replacement—validate all outputs with your own reasoning.

7. Visualize Insights

- Create charts that align with your narrative (markdown) using Matplotlib, Seaborn, and Plotly.
- Combine statistical insights with visual cues to tell a compelling story.
- Choose the right visuals—bar charts, line plots, scatter plots, heatmaps, etc.—based on your data.

8. Build a Dashboard

- Use Streamlit or Plotly Dash to design a clean, engaging dashboard for your insights.
- Include filters (sliders, dropdowns), key visuals, KPIs, and explanatory markdowns.
- Ensure your dashboard guides the viewer logically through your analysis journey.

9. Craft a Narrative and Structure Your Story

- Define your data story: What's the problem? What did the data reveal? What are the recommendations?
- Add section titles, summary notes, and context using markdown or Streamlit text components.
- Keep the narrative clear, business-focused, and aligned with your visuals.

10. Prepare and Deliver the Final Presentation

- Summarize your work in a 3–4 minute presentation using your dashboard and a structured PPT/report.
- Follow the Problem – Analysis – Insight – Recommendation structure.
- Highlight key visuals and explain your thinking clearly and confidently.

11. Peer Review and Feedback (Optional but Encouraged)

- Share your dashboard with peers or mentors for review.
- Discuss feedback and iterate to improve your visuals, storytelling, or summaries.

12. Submit Final Deliverables

- Your final submission should include:
 - Cleaned dataset (CSV or DB)
 - Python/Colab notebook or .py script
 - Final dashboard (Streamlit/Plotly Dash link or .py app)
 - PPT and PDF summary (with objectives, dataset details, key visuals, and insights).

13. Reflect and Showcase Your Work

- Reflect on what you learned and how you grew through this project.
- Optionally, post your dashboard or summary insights on LinkedIn, GitHub, or personal portfolio to signal your skills to future employers.

EDA Capstone Project Checklist

Use this checklist to stay organized and make sure you're on track at each stage of your project journey.

1. Topic Selection

- ☐ I selected a project topic from the provided list (or got mentor approval for a custom one).
- ☐ My topic allows for meaningful analysis and visual storytelling.

2. Data Collection

- ☐ I identified and downloaded a relevant dataset from a trusted source (Kaggle, govt site, etc.).
- ☐ My data is clean, ethically sourced, and sufficient for the chosen topic.

3. Data Import & Exploration

- ☐ I successfully loaded the dataset into Python (Colab or Jupyter).
- ☐ I explored the structure: column types, null values, value ranges, and inconsistencies.

4. Data Cleaning & Transformation

- ☐ I handled missing values, removed duplicates, and fixed data types.
- ☐ I renamed, merged/split columns, and filtered rows as needed.
- ☐ I stored the cleaned dataset in a suitable format (CSV/SQLite/MySQL).

5. Exploratory Data Analysis

- ☐ I performed univariate, bivariate, and multivariate analysis using Pandas/Seaborn/Matplotlib.

- ☐ I used grouping, aggregation, correlation, and pivot tables to extract insights.
- ☐ I identified interesting patterns, trends, and outliers.

6. GenAI Integration (Optional but Valuable)

- ☐ I used ChatGPT/Gemini to refine business questions or create markdown summaries.
- ☐ I validated the AI-generated summaries or visual explanations.

7. Data Visualization

- ☐ I created appropriate charts to support my analysis (bar, line, heatmap, scatter, etc.).
- ☐ My visuals are readable, labeled, and contextually placed.

8. Dashboard Creation

- ☐ I used Streamlit or Plotly Dash to build an interactive dashboard.
- ☐ The dashboard includes filters, summaries, and a logical flow.
- ☐ I included markdowns/titles to narrate the story clearly.

9. Narrative Crafting

- ☐ I created a coherent story: Problem → Analysis → Insights → Recommendation.
- ☐ I ensured all visuals supported my central message.

10. Final Presentation

- ☐ I prepared a 3–4 minute walkthrough of my dashboard and findings.
- ☐ I structured the presentation using a professional storyline.
- ☐ I rehearsed and can confidently explain my process and insights.

11. Documentation & Submission

- ☐ I compiled the following:
 - Cleaned dataset
 - Python notebook or script
 - Dashboard code/link
 - Project summary/report (PPT and PDF)
- ☐ All files are well-named and submitted as per instructions.

12. Reflection & Showcase

- ☐ I reflected on my project experience and key takeaways.
- ☐ I updated my portfolio, LinkedIn, or GitHub with this project (optional but recommended!).

Deliverables and Timelines

Below is a list of key deliverables you are expected to complete during your EDA Capstone Project. Each deliverable aligns with specific project milestones and showcases your applied learning.

#	Deliverable	Details	Timeline
1	One-Page Project Planning Document	Submit a one-page plan outlining your topic, business/data objective, key questions, dataset source(s), and your analysis approach. Mention Python libraries (Pandas, Seaborn, Matplotlib, etc.) and any GenAI tools (ChatGPT, Gemini) you may use. For group projects, specify team roles and your collaboration strategy.	Sprint 11
2	Cleaned & Transformed Dataset	Submit your dataset after cleaning and transformation using Pandas/NumPy. Include treatment for missing values, data type corrections, and column-level documentation. Attach a short data dictionary describing each column and its purpose.	Sprint 11
3	Exploratory Data Analysis Report	Perform univariate, bivariate, and multivariate analysis using Pandas, Seaborn, and Matplotlib. Present statistical summaries, trends, and patterns using both markdown and visuals. Submit your notebook (.ipynb/.py), a 2–3 page summary report (.PDF or .Docx) and presentation PPT (.pptx).	Sprint 12
4	GenAI-Aided Insight Summary	Use tools like ChatGPT or Gemini to refine your problem statement, generate markdown summaries, or build storylines. Include snapshots or summaries of how GenAI enhanced your insights or helped communicate your findings.	Sprint 12
5	Dashboard (Streamlit/Plotly Dash)	Create a visually clear, multi-section dashboard using Streamlit or Plotly Dash. Include charts, markdown annotations, and structured narratives. Ensure it aligns with best practices for layout, readability, and insight flow. Share code files.	Sprint 12
6	Presentation Deck	Prepare a slide deck for a 3–4 minute walkthrough of your project. Include topic, approach, key questions, methodology, insights, and visuals. Add screenshots of your dashboard and reflect on lessons learned. Save as PDF and PPT.	Sprint 14 and 15
7	Live Project Presentation	Deliver a short live presentation (3–4 minutes) using your dashboard and slides. Structure it using: Problem → Analysis → Insights → Conclusion → Recommendations. Be ready to answer peer or mentor questions confidently.	Sprint 14 and 15

Guidelines for Success

Make your EDA Capstone Project impactful, insightful, and presentation-ready by following these best practices:

- **Plan with Purpose:** Use a timeline or digital tracker (like Notion, Trello, or Excel) to schedule milestones—topic selection, dataset cleanup, EDA, dashboard creation, and presentation rehearsal. Make room for college deadlines and personal time.
- **Choose Credible Data Sources:** Pick datasets from **reliable, publicly available** platforms such as [Kaggle](#), [data.gov.in](#), [UCI Machine Learning Repository](#), or industry whitepapers. Avoid synthetic or unverifiable data.
- **Maintain Data Integrity:** Ensure your dataset is neutral and free of biases or sensitive themes. Focus on topics that allow objective, business-relevant analysis rather than political, religious, or controversial domains.
- **Collaborate Effectively (or Simulate It):** If working in a team, each member should be aware of and participate in all three key roles: Data Wrangler (cleans and prepares data), Insight Analyst (performs EDA and interprets findings), and Storyteller (creates dashboards and presentations). Regular check-ins are essential. If working solo, share drafts with peers or mentors to simulate collaboration and gather valuable feedback.
- **Stick to Python's Core Strengths:** Perform all cleaning and transformation tasks using NumPy and Pandas. Use Seaborn, Matplotlib, and Plotly for visuals. Document your transformation steps with markdowns/comments to keep your notebook well-organized.
- **Organize Your Work Using Cloud Tools:** Keep your project files (datasets, notebooks, dashboards, presentation decks) on Google Drive, GitHub, or OneDrive. Maintain a clean folder structure with versioning (v1, v2, final).
- **Document Everything:** Track: Libraries used, cleaning steps taken, visuals created and why, insights drawn, challenges faced. This will make writing your summary report and final deck much easier.
- **Master the Art of Visual Storytelling:** Make your dashboards clear, purposeful, and engaging. Use the right chart types for the right insights. Label clearly, remove clutter. Guide the viewer's eye through the story using markdown, color, and structure.
- **Integrate GenAI Thoughtfully:** Use ChatGPT or Gemini to brainstorm business questions, summarize insights, or generate explanatory markdown for charts. Don't copy blindly—validate and refine AI suggestions to suit your dataset and context.
- **Share and Seek Feedback:** Test your visualizations and story with peers. Ask: "Does this insight make sense to you?" "Is this chart easy to understand?" Incorporate their feedback early—don't wait till the end.
- **Rehearse and Refine:** Practice your 3–4 minute walkthrough multiple times. Keep it focused: Problem → Approach → Key Findings → Conclusion → Recommendations. Rehearsing boosts confidence and delivery flow.
- **Reflect and Keep Learning:** This project isn't just a submission—it's your first professional-grade data story. Embrace feedback, stay curious, and note what you'd

do differently next time. It all counts toward becoming a confident data analyst.

Evaluation Criteria

Given below are the criteria against which your project will get assessed. We encourage you to do a self-assessment. You can rate each of these parameters on a scale of 1-5 as per rating scale - 1: Not Met, 2: Partially Met, 3: Met, 4: Well Met, 5: Exceptionally Met.

Category	Criteria
1. Collaboration and Teamwork	1.1 Shared and organized project files, datasets, and codes effectively using GitHub/Drive.
	1.2 Worked collaboratively—communicated clearly and supported each other throughout the project.
2. Dataset Collection from Reliable Sources	2.1 Identified a credible and relevant dataset from public repositories (Kaggle, govt portals, etc.).
	2.2 Explained the dataset choice and aligned it with project goals.
3. Data Extraction, Cleaning, Transformation	3.1 Loaded data using Pandas and handled different file formats.
	3.2 Cleaned the dataset: removed duplicates, corrected types, handled missing values and outliers, etc. and stored cleaned data in a structured format (like SQLite/MySQL).
4. Data Analysis and Insight Extraction	4.1 Applied univariate, bivariate, and multivariate analysis techniques.
	4.2 Extracted meaningful insights using descriptive statistics, correlations, and group comparisons.
5. Visual Storytelling and Dashboard Creation	5.1 Designed clear and purposeful visuals using Matplotlib, Seaborn, Plotly.
	5.2 Created an organized dashboard using Streamlit/Plotly Dash to support insight story.
6. Project Documentation (Writing, Presentation & Narration)	6.1 Created a one-page plan with clear business questions and methodology.
	6.2 Designed a well-structured PPT/report following problem-solution-insight-benefit structure.
	6.3 Drafted a professional project summary using GenAI or markdown cells in Colab.
7. Presenting Data with Confidence	7.1 Delivered a 3–4 minute clear and engaging presentation with visuals.
	7.2 Responded to peer/mentor questions with clarity, demonstrating understanding of data story.

Project Topics

You may choose to explore one of the following topics:

Note: While one reference dataset has been suggested for each topic, learners are strongly encouraged to explore and collect their own datasets from reliable, real-world sources. This promotes research thinking, source evaluation, and data discovery skills—essential capabilities for any aspiring data analyst. Ensure the dataset you choose is publicly available, ethically sourced, and large enough to support meaningful exploration and storytelling.

1. Sales & Revenue Performance Analysis

- **Use Case:** Track sales growth, product performance, regional trends, profitability.
- **Real-World Relevance:** Core to business intelligence, FMCG, retail, and SaaS industries.
- **Data Access:** Retail Sales Data (Kaggle).

2. Banking Customer Churn Analysis

- **Use Case:** Identify trends and behaviors associated with customer churn.
- **Real-World Relevance:** Widely used in banking and telecom sectors for retention strategy
- **Data Access:** Bank Churn Dataset (Kaggle).

3. Marketing Campaign Effectiveness

- **Use Case:** Measure success of campaigns across segments, analyze conversion rates.
- **Real-World Relevance:** Essential for digital marketing, advertising, and e-commerce domains.
- **Data Access:** Marketing Campaign Dataset (UCI).

4. E-commerce Customer Purchase Behavior

- **Use Case:** Analyze buying patterns, basket size, frequency, and channel preferences.
- **Real-World Relevance:** Key for product recommendation, inventory, and UX improvement.
- **Data Access:** E-commerce Data (Kaggle).

5. Student Performance Analysis

- **Use Case:** Discover how demographic and institutional factors affect academic outcomes.
- **Real-World Relevance:** Critical in EdTech, academic institutions, and learning platforms.
- **Data Access:** Student Performance Data (UCI).

6. Customer Review Ratings & Trends

- **Use Case:** Analyze customer feedback, rating distributions, sentiment trends..
- **Real-World Relevance:** Used in product improvement and customer experience optimization.
- **Data Access:** Amazon Product Reviews (Kaggle)

7. Hospital Patient Health Data Analysis

- **Use Case:** Study admissions, treatment durations, cost patterns, and patient outcomes.
- **Real-World Relevance:** Important for public health analytics, hospital management.
- **Data Access:** Hospital Data (Kaggle).

8. Finance & Budgeting Analysis (Govt)

- **Use Case:** Explore allocation trends, budget vs actual, sector-wise funding.
- **Real-World Relevance:** Helps in policy-making, public administration, and civic tech.
- **Data Access:** Open Budget India – Data Portal or data.gov.in.

9. Electricity Consumption & Load Patterns

- **Use Case:** Examine usage peaks, regional load trends, and forecasting gaps.
- **Real-World Relevance:** Useful in energy analytics, utilities, and sustainable planning.
- **Data Access:** Household Power Consumption Dataset (UCI).

10. Supply Chain & Inventory Trends

- **Use Case:** Track product movement, reorder levels, delays, and cost efficiency.
- **Real-World Relevance:** Crucial in logistics, warehouse, retail, and operations management.
- **Data Access:** Supply Chain Dataset (Kaggle).

11. Air Travel and Tourism Trends

- **Use Case:** Analyze flight frequency, travel patterns, seasonal travel spikes.
- **Real-World Relevance:** Valuable in travel, hospitality, and tourism policy planning.
- **Data Access:** Indian Airline Performance (Kaggle).

12. Public Transportation Trends (Metro)

- **Use Case:** Monitor ridership data, ticket type usage, time slot analysis.
- **Real-World Relevance:** Key for urban transport planning and policy decisions.
- **Data Access:** Delhi Metro Ridership (data.gov.in).

13. App Store Usage & Ratings Analysis

- **Use Case:** Explore user behavior by app category, ratings, download trends.

- **Real-World Relevance:** Important for mobile product development and UI/UX design.
- **Data Access:** Google Play Store Apps (Kaggle).

14. IPL Cricket Performance Analytics

- **Use Case:** Analyze player stats, team performance, match outcomes, and fan trends.
- **Real-World Relevance:** High engagement sports dataset for predictive and performance analytics.
- **Data Access:** IPL Data (Kaggle).

15. Social Media Post Engagement

- **Use Case:** Discover engagement patterns, hashtag influence, post frequency impact.
- **Real-World Relevance:** Used in influencer marketing, campaign design, and digital PR.
- **Data Access:** Instagram Engagement Data (Kaggle).

Conclusion

This capstone project is your opportunity to bring together everything you've learned throughout the Exploratory Data Analysis using Python course. It moves you beyond theoretical understanding into real-world practice—empowering you to transform messy, raw datasets into actionable insights using Python's powerful data analysis tools.

By choosing a topic of personal or professional interest, you will apply and reinforce your skills by:

- Framing business questions that guide your analysis journey,
- Extracting, cleaning, and transforming datasets using Pandas and NumPy,
- Conducting univariate, bivariate, and multivariate analyses using Matplotlib, Seaborn, and Plotly,
- Leveraging GenAI tools like ChatGPT or Gemini to refine your approach and summarize findings,
- Creating interactive visual summaries or dashboards using Streamlit or Plotly Dash,
- Presenting a compelling data story with clarity, confidence, and purpose.

This project is more than just an academic milestone—it's a simulation of the real work data analysts do every day. It helps you:

- Sharpen your critical thinking and problem-solving abilities,
- Improve storytelling skills with data and visuals,
- Present insights in a structured, stakeholder-friendly format,

- Build a strong, portfolio-ready project that signals your readiness for internships, placements, or freelance opportunities.

Ultimately, this is your stage to showcase your **data literacy, creativity, and Python-powered analytical thinking**—skills that are in high demand across industries. Own your process, trust your insights, and enjoy the experience of making data meaningful.